

Thursday 12-9-25

0-335 • Wk 05

Find outputs (HOMEWORK)

def square(fun): # fun is num

def inner1():

x = fun() # $x = \text{num}() \rightarrow x = 10$

return x * x

return inner1

def double(fun): # fun is inner1

def inner2():

y = fun() # $y = \text{inner1}() \rightarrow y = 100$

return 2 * y

return inner2

@double - # $\text{num} = \text{double}(\text{square}(\text{num})) \rightarrow \text{num} =$
 $\text{double}(\text{inner1}) \rightarrow \text{num} = \text{inner2}$

@square

def num():

return 10

end of the function

print(num()) # print(inner2) $\rightarrow \text{print}(200)$

Find outputs (HOMEWORK)

def bold(fun): # fun is inner2

def inner1():

return '' + fun() + '' # return '' + inner2()

return inner1

def italic(fun): # fun is inner3

def inner2():


```

return '<i>' + func() + '</i>'
return inner2
def underline(func): # func is f1
    def inner3():
        return '<u>' + func() + '</u>'
    return inner3

```

@bold · # f1 = bold(italic(underline(f1))) →
 f1 = bold(italic(inner3)) → f1 = bold(inner2)
 → f1 = inner1 → Ref f1 points to inner1

@italic :

@underline

```
def f1():
```

```
    return 'Hello World'
```

End of the function

```
print(f1()) # print(inner1())
```

```
# <b><i><u>Hello World</u></i></b>
```